Getting started

A practical guide to preparing rainfall data, running an intensity–duration–frequency assessment and producing a traceable technical report.

The Atlas supports screening and hydrologic analysis. A generated report records the active analysis; it does not certify a design or establish a regulatory return period.

Five-step workflow

• Choose data. Upload a local gauge workbook, retrieve online precipitation, or select a CMIP6 projection workflow.

• Confirm location and time coverage. Check coordinates, spatial extraction and native temporal interval.

• Review quality control. Resolve rejected dates, duplicates, units and incomplete years before interpreting a fit.

• Fit and compare. Select stationary or nonstationary analysis and inspect diagnostics and alternative distributions.

• Export an evidence package. Save the report, Excel calculations and chart together.

Before you begin

Use a modern browser. Online retrieval requires internet access and a reachable data provider. Local gauge processing runs in the browser. Keep a backup of the original workbook and record its source, station name, coordinates and units.

Navigation

Data selects the source; Map controls spatial selection; Analysis contains fitted curves and diagnostics; Report contains the generated document; Manual opens this guide.

Prepare and upload gauge data

Input preparation

Use Excel or CSV with a dated precipitation series. Each timestamp should describe one accumulation interval, and rainfall values should be numeric. Clearly identify dates, precipitation, units and whether the native interval is daily, hourly or sub-hourly. Compatible worksheets are scanned automatically; inspect the import summary rather than assuming a particular sheet was selected.

Check	What to do
Dates and times	Use unambiguous dates, preferably ISO format. Confirm timestamp convention and interval boundaries with the source.
Precipitation	Distinguish zero rainfall from missing observations. Do not replace missing values with zero.
Units	Confirm accumulation depth in millimetres or the supported unit selected in the interface. Do not treat a rate as an interval total.

Check	What to do
Duplicates	Investigate repeated timestamps in the source. Check rejection and acceptance counts after import.
Location	Enter a meaningful gauge name and coordinates where available. Gauge fitting does not use district boundaries.

After uploading

• Read the source label, detected native interval, accepted date range and quality-control counts.

• Compare accepted records with your source workbook.

• Inspect retained years and annual maxima. At least 90% coverage is required for a retained year.

• If a result is unavailable, correct the underlying file or select a suitable model; do not fill missing observations solely to enable fitting.

Stationary fitting requires at least eight eligible annual maxima. Fewer than 30 is provisional. Nonstationary fitting requires at least 20 eligible or covariate-matched maxima. These minimums are software eligibility rules, not proof of reliability.

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Retrieve online precipitation

Choose the source and native interval

The interface offers ERA5, ECMWF IFS, ERA5-Land, CHIRPS, NASA POWER and supported IMERG workflows. Availability depends on provider, period and interval. CHIRPS is daily; authenticated IMERG half-hourly archive access is not connected. Read the source capability message before retrieval.

ECMWF IFS: hourly or daily precipitation is available from 2017 through the explicit Open-Meteo ecmwf_ifs archive. It is an operational model-analysis product, not a homogeneous long-term reanalysis. Model upgrades and its short record require gauge validation and caution for rare return periods. IFS shares Open-Meteo quotas with ERA5. District values use a representative request mesh mapped to model cells and are approximate.

Point or district extraction

• Select a point/grid-cell series or a supported district spatial series.

• For a point, verify latitude and longitude. For a district, verify the selected boundary and extraction method.

• Select the requested dates and interval, then retrieve precipitation.

• Wait for successful completion. Check returned dates, record count, units, provider and spatial treatment.

• Inspect the precipitation plot and quality-control summary before fitting.

A gridded point is not a rain gauge. District sampling or polygon statistics are not identical to a single grid cell. Retain the spatial method in any comparison or report.

If retrieval fails

Read the provider error. Try a shorter time window or supported interval when the request is too large. Check coordinate and date limits. If an alternate provider is used, verify that the report identifies the actual

dataset and interval. A failed or pending request must not be interpreted as a zero-rainfall record.

Provider, dataset name, retrieval time, requested and returned period, precipitation units, point coordinates or boundary, spatial extraction method and accepted record count.

Daily versus hourly

Daily input directly supports 24-hour totals. Native hourly data support rolling 1-, 2-, 3-, 6-, 12- and 24-hour durations. Shorter durations require the stated derivation and local validation. A daily series cannot become an observed hourly series by selecting a shorter chart duration.

Fit and interpret IDF curves

Stationary analysis

Select a supported distribution or curve method and compare results. A stationary model holds its parameters constant over time. Review stationarity diagnostics, fit diagnostics, method sensitivity and the number of eligible annual maxima. Failure to reject a trend is not proof that future rainfall will be stationary.

Nonstationary analysis

Choose a time-varying GEV or a supported annual covariate. Confirm at least 20 eligible matched annual maxima. Record the evaluation year or covariate value, whether it is interpolation or extrapolation, the model comparison and the preferred model decision. Correlation alone is not model-selection evidence.

Calculate intensity and understand its constants

Above the chart, enter t in minutes and choose return period T. Intensity i in mm/hour and depth update automatically within the fitted range. Expand “What do the equation and constants mean?” for an explanation. The default Sherman calculator uses i=a*T^m/(t+b)^n. It estimates b only with at least three independently supported native durations; daily-derived curves retain an explicitly labelled b=0 simplification. Boundary solutions require review. The reported bounds restrict evaluation. Select Power · Excel-style trendline to use i=c×t^p: c is a unit-dependent fitted scale and p is the fitted duration exponent. The website regresses ln(i) on ln(t) for each return period, using the active IDF values. R^2 describes that log-space fit and can equal one when duration scaling imposed a power relationship.

Report revision 4.0 includes current c and p values, a worked substitution, fixed versus fitted constant definitions, calculation steps and an acronym list. Appendix A explains formulas; Appendix E defines acronyms. A general formula description does not imply that a method was applied to the active data.

Read the chart and tables

Quantity	Interpretation
Intensity, mm/hour	Mean rainfall rate over the stated duration.
Depth, mm	Intensity × duration in hours. At 24 hours, depth = 24 × intensity.
T-year return level	For a stationary model, annual exceedance probability is 1/T; a 100-year level has 1% AEP.

Quantity	Interpretation
Nonstationary return level	Conditional on the stated year or covariate value; it is not a timeless recurrence interval.
95% bootstrap interval	Sampling uncertainty under the fitted workflow, not all uncertainty.

Engineering interpretation

Use the governing authority’s criteria to choose the design return period. For a time of concentration calculation, verify flow-path length, slope units and geometric method; a calculated duration does not establish rainfall validity at that duration.

The 250- and 500-year levels extend far beyond most available records. Check uncertainty, regional evidence and sensitivity to distribution choice before design use.

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CMIP6 projection workflow

Retrieve and correct daily model data

• Select a GCM, realization/member, SSP, period and spatial request.

• Retrieve the NASA NEX-GDDP-CMIP6 daily precipitation where the subset service is reachable.

• Review calendar treatment, precipitation conversion to mm/day and complete model years.

• Retrieve the historical GCM and the selected gauge or ERA5 reference for the 1985–2014 correction baseline.

• Apply monthly multiplicative quantile delta mapping (QDM) before extracting future annual maxima.

• Fit each model separately and review the ensemble of model-specific return levels.

ERA5 correction must be described as reanalysis-corrected. It is not gauge correction. Do not pool model-years as independent observations. Native daily GCM data directly support only 24 hours; observed hourly duration ratios are needed for the supported sub-daily correction workflow.

Correct a district spatial series

Import corrected results when daily retrieval is unavailable

Choose Import bias-corrected results workbook. Include Read Me rows for Location, Reference and Historical overlap; a Validation sheet with result, evidence, severity, impact and required action; and a Model IDF sheet with model, member, scenario, period, return period, Primary_Corrected_24h_Depth_mm and Primary_Corrected_Intensity_mm_hr.

An optional Ensemble IDF sheet is reconciled against model rows. At least three models, common return periods, valid coordinates and labels, increasing curves, and consistent depth/intensity values are required. Preserve workbook caveats after import.

P10–P90 is inter-model spread, not a 95% sampling-confidence interval. An imported summary supports 24-hour return levels only and does not reconstruct daily records, annual maxima, stationarity tests or alternative distribution fits.

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Download and review the report

Available downloads

Format	Use
Print / Save PDF	Open browser printing, select Save as PDF and A4. Use 100% scale; enable background graphics where available. Turn off browser headers/footers if they duplicate the report's own.
Word report (.docx)	A native editable Word document with headings, tables, chart image and page numbering. Pagination may vary between Word and LibreOffice.
Editable HTML	A self-contained document with embedded styles and vector chart. Open in a browser to inspect or print later.
Excel workbook	Calculation and provenance evidence. Retain alongside the report.
User manual	This guide is available in PDF and browser-readable HTML.

Review before sharing

- Confirm the cover location, source, model, method and issue date.
- Read the assessment status. A report generated before data loading contains no fitted results.
- Check the executive decision table against the full IDF table and Excel output.
- Verify measured versus derived durations, units and the selected evaluation year or covariate.
- Read quality-control exclusions, uncertainty, limitations and source caveats.
- Inspect every PDF page for clipped tables, missing charts and isolated headings.
- Retain the original input and exported evidence. Record any manual edits separately.

Recommended delivery package

Include the PDF report, editable DOCX or HTML, Excel calculation workbook, chart, source-data citation and this manual. Name files with the location, source, scenario where applicable and issue date. Ask an independent reviewer to record their review outside the automatically generated analysis.

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Troubleshooting and reproducibility

Symptom	Action
No fitted result	Confirm successful import/retrieval, eligible annual maxima, selected source and model requirements. Re-export after fitting.
Many rejected records	Inspect timestamps, nonnumeric values, negative rainfall, unit selection and duplicate intervals in the original file.
Sub-daily output unavailable	Confirm native interval. Imported CMIP6 summary curves support 24 hours only; use an eligible hourly workflow where appropriate.
Nonstationary fit unavailable	Check at least 20 eligible maxima and matched covariate years. Use the model-status message to identify the missing requirement.

Symptom	Action
Unexpectedly large return levels	Check units, outliers, shape sensitivity, record length and rare-event extrapolation. Compare supported distributions.
Word export fails	Read the export error, retry in a modern browser, or save HTML/PDF. Allow the download to finish before closing the tab.
PDF table is cut off	Use A4 and 100% scale. Inspect print preview and disable custom browser scaling. HTML retains horizontal scrolling on small screens.

Reproducibility record

Retain the repository commit/version, source workbook, source and retrieval details, selected location and period, QC summary, distribution, stationary/nonstationary setting, covariate evaluation, duration derivation and exports. A report identifier describes the selection; it is not a cryptographic dataset fingerprint.

Daily duration scaling: Pakistan literature preset

Select Daily CMIP6 projections, retrieve and bias-correct daily rainfall or import corrected daily return levels. The default duration method is Pakistan literature · IMD one-third (Haseeb et al., 2025).

Verified source: Haseeb, F., Ali, S., Ahmed, N., Alkhuraiji, W. S., Đurin, B., and Youssef, Y. M. (2025). Forecasting Rainfall IDF Curves Using Ground Data and Downscaled Climate Projections to Enhance Flood Management in Punjab, Pakistan. Atmosphere 16(11), 1271. DOI: 10.3390/atmos16111271. Section 3.2, Equation (1), explicitly uses the IMD reduction formula with a depth exponent of exactly one-third.

The implemented equation is $P_t = P_{24}(t/1440)^{(1/3)}$; $i_t = 60P_t/t$. Equivalently, $i_t = i_{24}(t/1440)^{(-2/3)}$. The preset uses only the paper's durations: 1, 2, 3, 4, 6, 12 and 24 hours. The calculator is restricted to this 60–1440-minute range. The 24-hour depth is preserved. No sub-hour rainfall is inferred in this preset.

One-third is a published empirical selection, not an exponent estimated from the selected district's data. The cited study concerns Punjab cities. Applying its method to another district or to district-average rainfall is a transfer requiring local validation. The website adopts its duration-reduction component; it does not claim to reproduce the paper's entire EQM bias-correction, climate-model-selection or distribution-fitting workflow.

Use the Duration scaling method selector to choose Custom exponent only when you have a defensible separate basis. This mode enables editable α and derived 5-minute to 24-hour values; sub-hour values are unvalidated extrapolations. Disable the scaling checkbox for 24-hour-only results. Reports and the Excel Duration Scaling sheet retain the method, exact exponent, durations, DOI and limitations. Confidence bounds propagate daily uncertainty only, excluding scaling uncertainty. A perfect power-fit R^2 follows from the imposed equation and is not validation.

Intensity units: mm/hr and inch/hr

The calculator shows both units together. Separate chart panels show mm/hr and inch/hr, each with its own axis, equations and image download. Divide mm/hr by exactly 25.4 to get inch/hr. Excel includes both intensity units and converted uncertainty bounds; the technical report includes a dual-unit results table. Equation coefficients a or c must be divided by 25.4 for inch/hr; duration, offsets, exponents and return periods stay unchanged. Internal fitting and rainfall depths remain metric.

Select curves and inspect values

In either unit panel, click a legend entry to hide or show its return-period curve. Click a curve to focus it; use Clear curve selection or Show all curves to reset. Move the cursor within the plot to see the nearest plotted duration, return period and rainfall intensity in that panel's units. Values are plotted data points, not additional interpolated estimates. Touch a point, or focus it with Tab and use Enter/Space, to inspect it. Single-duration frequency plots also show cursor values.

Permanent default: uploaded Districts.zip

The website permanently bundles the supplied Districts.zip map and loads it automatically on each visit. Choose Province / territory, then District / source unit, or click a polygon. The map contains 161 named source units, including the supplied combined Kashmir polygon and historical FATA grouping. The source does not provide the newer district subdivisions from the previous map, and these are not invented or mixed into the uploaded map.

The horizontal coordinate system is WGS 84 (EPSG:4326). The source's vertical metadata are not used for this two-dimensional map. Coordinates were not reprojected, rounded, simplified or clipped. Two invalid self-intersecting rings, in Barkhan and Batagram, required topology repair; their relative area changes are below 10^{-12}. The original ZIP and a conversion/repair manifest are retained and downloadable alongside the active GeoJSON.

The same selected polygon is used by the existing rainfall extraction and bias-correction workflows. Unique source IDs use the archive checksum and record number, because the original OBJECTID field repeats for some records. Reports and exports identify the user-provided map and archive version. Dataset year and license were not supplied in the archive. The source's classifications and disputed extents do not determine current administrative status or sovereignty. The optional 2017 tehsil layer remains a separate dataset.

Retrieved precipitation on the district map

Retrieve a district series for the selected boundary, then use the precipitation map beneath the main boundary map. Choose a date/hour or move the time slider. The default view colours the polygon with its retrieved district-series depth in mm and inches, using a fixed series colour scale. It does not imply spatially uniform rain or represent an IDF return level.

For ERA5, IFS and CMIP6, choose Raw source grid / sampling locations and retrieve the selected interval. Hover, focus or select a location to inspect its coordinates and precipitation. ERA5 uses representative sampling, not a complete raster; IFS shows returned locations without invented footprints. CMIP6 uses valid source cells selected by the existing extraction rule and identifies nearest-cell fallback. Raw source cells are not independently bias-corrected: district QDM applies to the district series only. Missing cells/dates are not filled or interpolated.

Use Download displayed precipitation CSV or Download spatial map SVG to retain values and source/boundary provenance. Other providers show their retrieved district aggregate when cell fields are unavailable. Point/gauge records cannot be displayed as district rainfall fields. New boundary or data selections invalidate the old map data.

Average annual rainfall for the retrieved period

The selected-district map panel shows average annual rainfall in mm/year and inches/year, the full-period retrieved total and annual coverage. Each annual total is the sum of daily or hourly rainfall depths; the mean is the arithmetic average of complete annual totals. It is not an annual maximum or IDF intensity.

Years must have 100% of their source-calendar intervals for this total-rainfall statistic. Partial years and wholly missing years are listed and excluded from the mean, without gap filling or annualization. Their available rainfall remains part of the separately displayed full-period retrieved total. When every year is complete, the mean covers the entire period; otherwise it is labelled as the complete-year subset.

Gregorian leap years and 365-day model calendars are handled separately. Preceding-hour records are aligned to interval starts for annual assignment; a final year can require the following 1 January midnight record.

Select Average annual rainfall in Map display to map the period mean. Date controls are inactive for that view. Download annual totals and coverage CSV for the calculation audit; the main Excel export includes an Annual Rainfall sheet, and the report executive summary states the same mean and coverage. Corrected district series are used where active; no cell-wise annual field is inferred from a raw daily grid snapshot.

Developers

The repository is saqibmahmood7117/pakistan-idf-atlas. Use Node.js 22.13 or newer, run npm ci, then npm run dev. Validate changes with npm test, npm run lint and npx tsc --noEmit. Report exports are implemented in lib/report-export.ts and the report section of app/page.tsx.

Manual revision 12.0 · November 2026. Interface and provider availability may evolve. The active application’s metadata and validation messages remain the authority for a particular analysis.